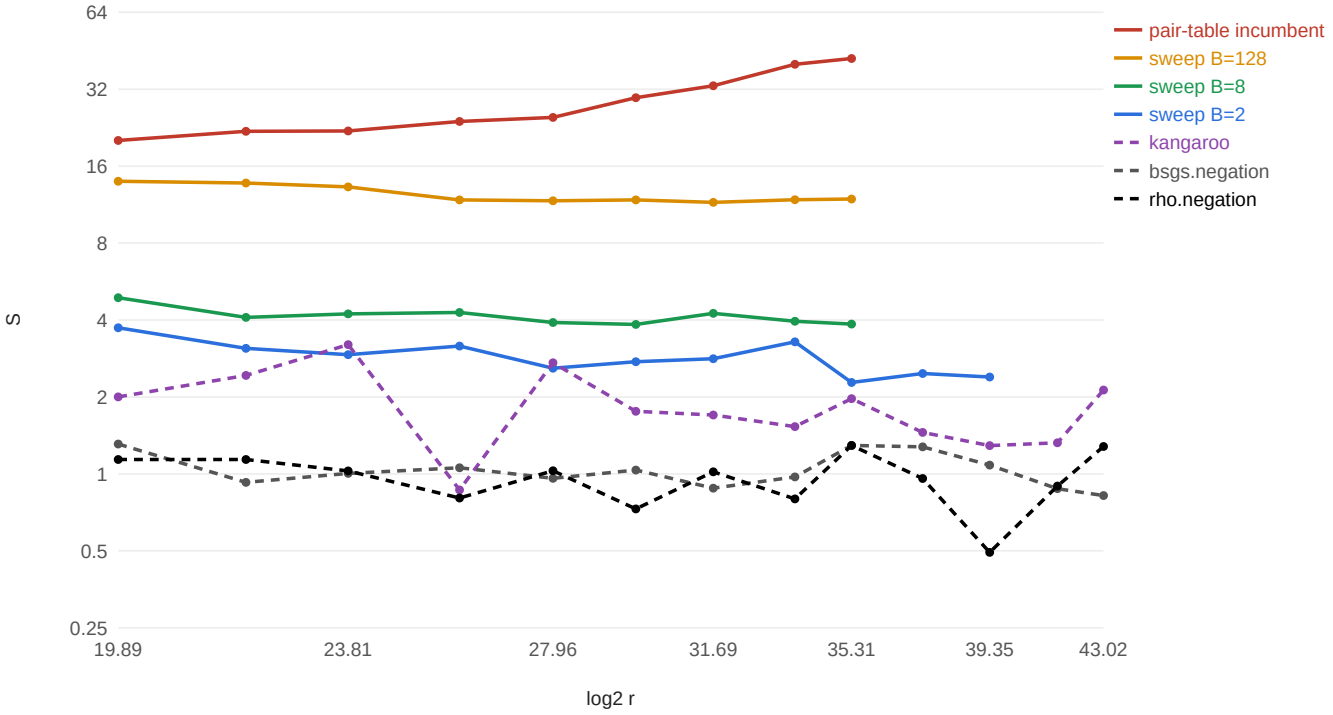
The residual sweep, exhaustively (2026-10-28)

Lead `ICP-residual-partial-relations`. Sizes 20 to 44 bits, three sessions each, the sweep against the generic searches and the pair-table incumbent, with variants (no sign fold, distinguished points, pair-table hybrid) and base-shape controls.

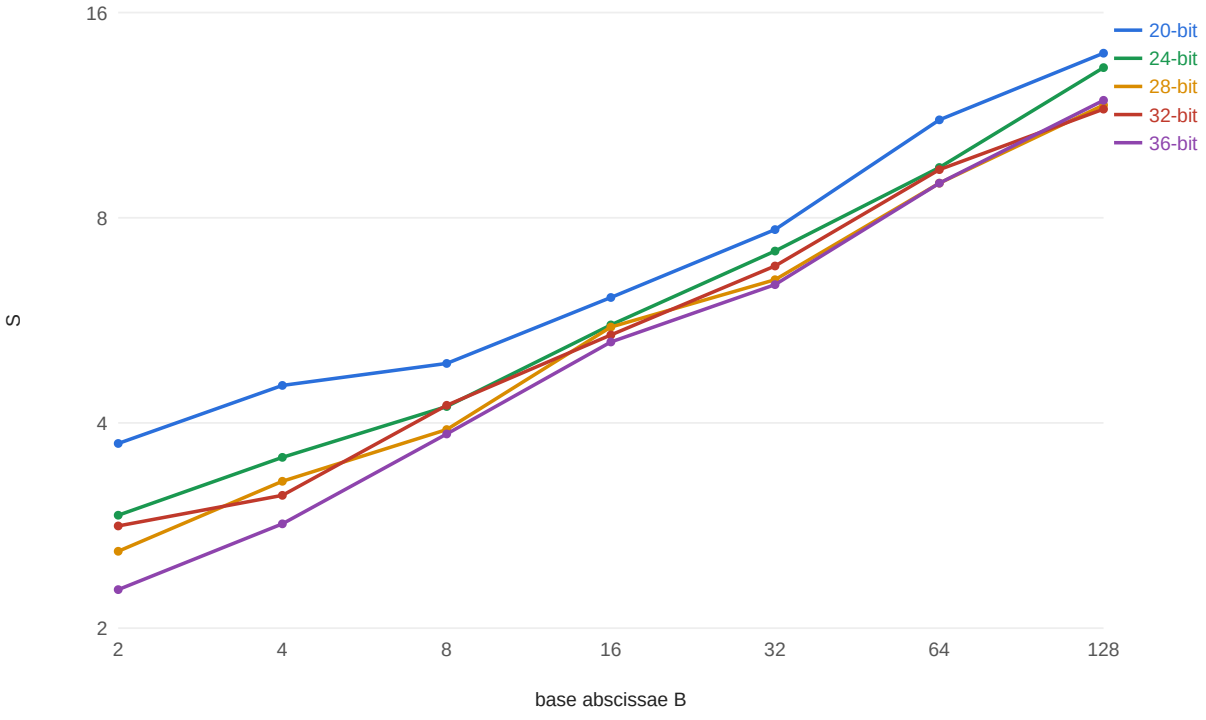
Counted S against curve size (log2 r)

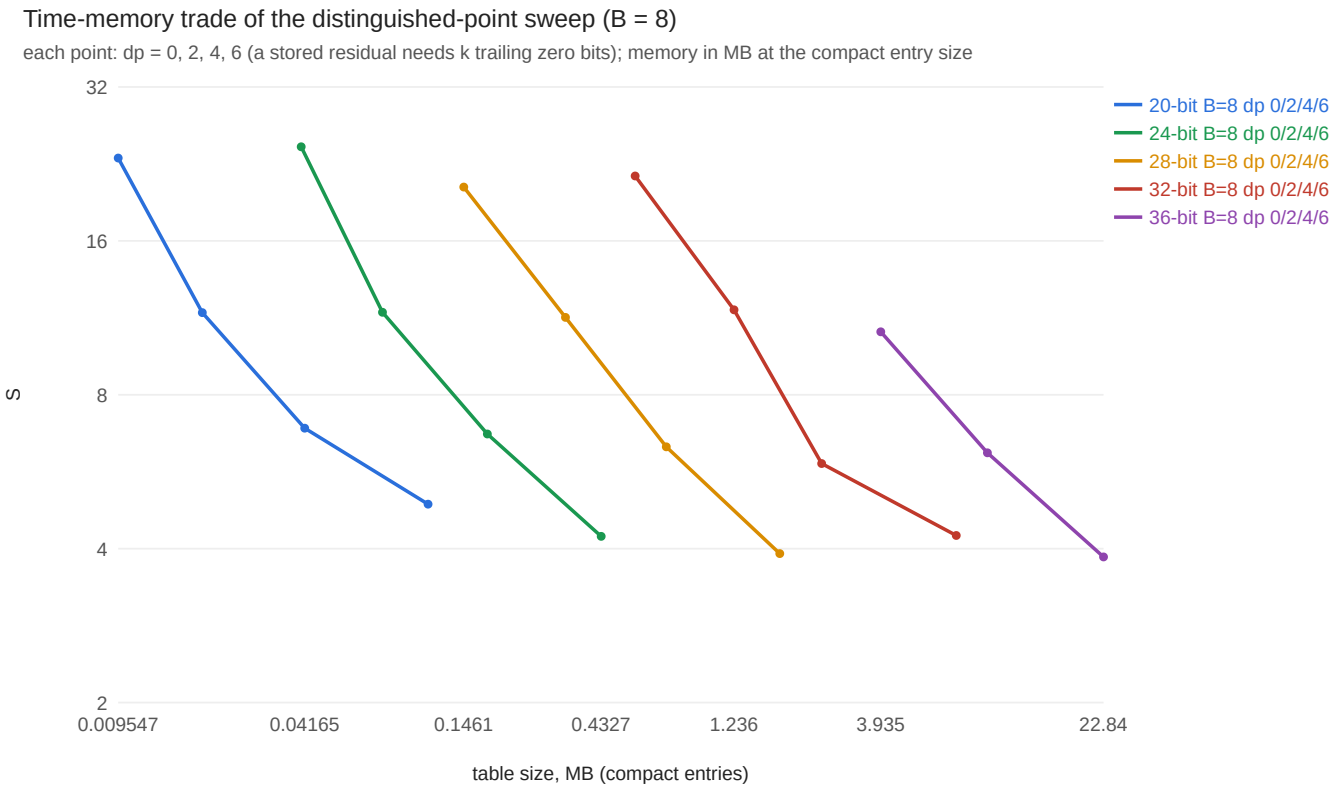
sweep at fixed base B, the planner-tuned pair-table incumbent, and the generic searches



Counted S of the sweep against its base size B

model: S = c * sqrt(2 B), slope 1/2 in log-log





Findings

- **What it is.** A birthday collision search that runs through the index-calculus pipeline. Evidence: (1) at a fixed base the counted S does not grow with the curve (slope of log2 S against log2 r, 95% bootstrap interval: B=2 -0.025 [-0.034, -0.017], B=8 -0.014 [-0.020, -0.009], B=32 -0.019 [-0.023, -0.014], B=128 -0.017 [-0.021, -0.013]), exactly like rho.negation -0.010 [-0.022, +0.001], bsgs.negation -0.007 [-0.013, -0.002] and kangaroo -0.023 [-0.031, -0.016]; the tuned pair-table incumbent grows: +0.071 [+0.069, +0.074]. (2) The collision rows found equal the birthday count $N^2/(r\,2^k)$ in every variant (ratios 0.7 to 1.2, table below). (3) Replacing the smallest abscissae by pseudo-random strided ones changes S by 1.03 [1.00, 1.07]: the base has no index-calculus structure to contribute. (4) Adding the pair table changes S by 1.00 [1.00, 1.00]. (5) With a known-log base a single collision row pins the logarithm (rows used = 1): that is a baby-step/giant-step-style search (S ratio to the abscissa base: progression 0.65 [0.61, 0.69], random scalars 0.35 [0.33, 0.37]).
- **Cost model.** A target contributes 2B signed residuals, one group addition each. After N such additions the expected number of colliding pairs is N^2/r (classes {Z, -Z}: $N^2/(2\,r/2)$), without the sign fold $N^2/(2r)$, and $N^2/(r\,2^k)$ when only residuals whose hashed abscissa has k trailing zero bits are kept (equal abscissae hash alike, so a pair collides only if it passes the filter). The solve stops when the column of d is pinned, which takes K collision rows: K = 1.12 B + 4.0 for the abscissa base (least squares over sizes 20 to 36), 51% of the rows linearly dependent, and K = 1 for a known-log base. So N = sqrt(K r) and S = N/sqrt(r) is about c sqrt(K): flat in r, growing like sqrt(B) once B exceeds a few, smallest at the smallest base. The table holds N entries ($N/2^k$ with the filter).
- **Placement at 36 bits.** rho.negation 1.29, rho.negation_lanes 1.33, bsgs.negation 1.29, kangaroo 1.97; the sweep is 2.28 at B=2, 3.85 at B=8 and 11.89 at B=128; the tuned pair-table incumbent is 42.2. The generic searches stay flat to 44 bits and the plain B = 2 sweep to 40 bits (past that only the limited-memory variants run: memory). The sweep sits between the generic searches and the incumbent, at the generic searches' exponent.
- **Distinguished points (B = 8, paired S against the plain sweep, pooled over sizes).** k=2 1.44 [1.40, 1.48], k=4 2.57 [2.48, 2.66], k=6 5.01 [4.76, 5.28]. Stored residuals fall by only 2.7, 5.9 and 11.9 times (not 4, 16, 64: the run lasts longer), while S grows 1.45, 2.64 and 5.17 times, so S x memory falls to 0.53, 0.45 and 0.43 of the plain sweep's and then flattens. The trade never reaches rho's constant memory at a rho-like S.
- **Memory (table entries, 36 bits).** sweep B=2 375346, B=8 748282; bsgs.negation 103364 baby steps; kangaroo 1562 distinguished points; rho.negation none; the pair-table incumbent 4194304. The counted unit does not see any of it. The small-base sweep needs several times BSGS's memory for a larger S, which is the sense in which BSGS dominates it.
- **Verdict.** Not a genuine index calculus: its exponent is the generic one and the algebraic structure of the base contributes nothing measurable. It is a real, verified and cheap arm of the pipeline (the lowest S of any abscissa-base arm we have measured; the known-log bases are lower still and are collision searches by construction), but under the owner's rule that relabelling is not an improvement it does not go on the leaderboard; the owner decides.

Birthday model check (collision rows found against N^2 fold / $(2\,r\,2^k)$)

bits	arm	observed	predicted	ratio
24	b8-klrand	1.9	1.5	1.25
24	b8-klprog	6.0	8.4	0.71
24	b8-null-strided	14.3	18.4	0.78
24	b8-nofold	14.8	19.1	0.77
24	b8-dp6	7.9	8.5	0.93

24	b8-dp4	7.5	7.6	0.99
24	b8-dp2	8.6	9.6	0.89
24	sweep-b128	166.6	170.8	0.98
24	sweep-b64	78.3	84.4	0.93
24	sweep-b32	42.2	47.7	0.88
24	sweep-b16	24.2	27.0	0.90
24	sweep-b8	12.2	14.5	0.84
24	sweep-b4	6.6	9.2	0.72
24	sweep-b2	4.4	4.7	0.94
32	b8-klrand	1.6	2.3	0.68
32	b8-klprog	7.9	10.1	0.78
32	b8-null-strided	13.3	13.5	0.99
32	b8-nofold	14.3	14.6	0.98
32	b8-dp6	5.7	6.8	0.83
32	b8-dp4	7.0	8.0	0.88
32	b8-dp2	7.9	7.9	1.01
32	sweep-b128	134.3	136.5	0.98
32	sweep-b64	80.7	88.9	0.91
32	sweep-b32	43.1	45.8	0.94
32	sweep-b16	26.2	27.6	0.95
32	sweep-b8	13.8	16.2	0.85
32	sweep-b4	9.2	8.1	1.14
32	sweep-b2	4.7	5.6	0.84
36	b8-klrand	2.0	1.5	1.37
36	b8-klprog	9.4	11.0	0.86
36	b8-null-strided	15.5	18.0	0.86
36	b8-nofold	11.9	12.2	0.97
36	b8-dp4	5.9	6.7	0.89
36	b8-dp2	8.4	8.8	0.96
36	sweep-b128	141.9	142.1	1.00
36	sweep-b64	74.2	80.8	0.92
36	sweep-b32	40.3	40.1	1.00
36	sweep-b16	24.3	27.4	0.89
36	sweep-b8	14.5	13.4	1.08
36	sweep-b4	6.8	6.8	1.01
36	sweep-b2	3.9	3.6	1.09

Variants at B = 8, paired S against the plain sweep

variant	pooled ratio	95% interval	n
strided pseudo-random abscissae (null control)	1.034	[1.000, 1.069]	324
no sign fold	1.331	[1.294, 1.368]	324
pair table added (hybrid)	1.004	[1.003, 1.004]	324
known-log base, progression	0.653	[0.615, 0.694]	324
known-log base, random scalars	0.352	[0.332, 0.374]	324
distinguished points, k = 2	1.438	[1.398, 1.479]	324
distinguished points, k = 4	2.570	[2.481, 2.662]	324
distinguished points, k = 6	5.012	[4.758, 5.280]	252

All arms

bits	arm	verified	mean S	S / rho.negation	paired S vs incumbent	table entries	table MB (compact)
20	pair-b16-b256	36/36	69.73 ± 0.03	61.3	3.461	65536	1.00
20	pair-b16-b128	36/36	21.29 ± 0.09	18.7	1.056	16384	0.25

20	pair-b16-b64	36/36	10.78 ± 0.15	9.5	0.533	4096	0.06
20	pair-walk-b128	36/36	20.17 ± 0.16	17.7		16384	0.25
20	b8-klrand	36/36	2.53 ± 0.12	2.2	0.121	1234	0.04
20	b8-klprog	36/36	3.46 ± 0.23	3.0	0.158	2283	0.07
20	b8-null-strided	36/36	5.12 ± 0.13	4.5	0.251	3823	0.12
20	b8-hybrid	36/36	4.96 ± 0.16	4.4	0.242	3613	0.11
20	b8-nofold	36/36	5.86 ± 0.24	5.2	0.282	4515	0.14
20	b8-dp6	36/36	23.23 ± 1.01	20.4	1.116	313	0.01
20	b8-dp4	36/36	11.58 ± 0.51	10.2	0.552	608	0.02
20	b8-dp2	36/36	6.88 ± 0.19	6.0	0.337	1365	0.04
20	sweep-b128	36/36	13.95 ± 0.26	12.3	0.688	12571	0.38
20	sweep-b64	36/36	11.13 ± 0.18	9.8	0.550	9839	0.30
20	sweep-b32	36/36	7.69 ± 0.14	6.8	0.379	6465	0.20
20	sweep-b16	36/36	6.11 ± 0.13	5.4	0.301	4876	0.15
20	sweep-b8	36/36	4.89 ± 0.16	4.3	0.238	3613	0.11
20	sweep-b4	36/36	4.54 ± 0.15	4.0	0.220	3101	0.09
20	sweep-b2	36/36	3.73 ± 0.16	3.3	0.180	2158	0.07
20	kangaroo	36/36	2.00 ± 0.13	1.8	0.089	112	0.00
20	bsgs-neg	36/36	1.31 ± 0.02	1.2	0.065	494	0.01
20	rho-lanes16	36/36	1.40 ± 0.10	1.2	0.063	0	0.00
20	rho-neg	36/36	1.14 ± 0.11	1.0	0.048	0	0.00
22	pair-b16-b384	36/36	75.15 ± 0.08	66.0	3.436	147456	2.25
22	pair-b16-b192	36/36	22.11 ± 0.13	19.4	1.010	36864	0.56
22	pair-b16-b96	36/36	12.05 ± 0.21	10.6	0.548	9216	0.14
22	pair-walk-b192	36/36	21.88 ± 0.12	19.2		36864	0.56
22	b8-klrand	36/36	1.89 ± 0.06	1.7	0.085	2389	0.07
22	b8-klprog	36/36	2.68 ± 0.19	2.4	0.111	4114	0.13
22	b8-null-strided	36/36	4.21 ± 0.22	3.7	0.182	7043	0.21
22	b8-hybrid	36/36	4.13 ± 0.18	3.6	0.182	6829	0.21
22	b8-nofold	36/36	6.00 ± 0.18	5.3	0.270	10472	0.32
22	b8-dp6	36/36	19.12 ± 1.35	16.8	0.753	546	0.02
22	b8-dp4	36/36	10.19 ± 0.63	9.0	0.427	1167	0.04
22	b8-dp2	36/36	5.61 ± 0.30	4.9	0.240	2426	0.07
22	sweep-b128	36/36	13.74 ± 0.29	12.1	0.623	26609	0.81
22	sweep-b64	36/36	10.00 ± 0.11	8.8	0.456	19024	0.58
22	sweep-b32	36/36	7.65 ± 0.18	6.7	0.346	14219	0.43
22	sweep-b16	36/36	5.86 ± 0.19	5.2	0.262	10497	0.32
22	sweep-b8	36/36	4.10 ± 0.18	3.6	0.180	6829	0.21
22	sweep-b4	36/36	3.17 ± 0.16	2.8	0.137	4769	0.15
22	sweep-b2	36/36	3.10 ± 0.23	2.7	0.126	4170	0.13
22	kangaroo	36/36	2.43 ± 0.12	2.1	0.106	300	0.01
22	bsgs-neg	36/36	0.93 ± 0.04	0.8	0.041	1018	0.02
22	rho-lanes16	36/36	1.45 ± 0.08	1.3	0.062	0	0.00
22	rho-neg	36/36	1.14 ± 0.09	1.0	0.046	0	0.00
24	pair-b16-b512	36/36	71.20 ± 0.07	69.3	3.246	262144	4.00
24	pair-b16-b256	36/36	22.07 ± 0.09	21.5	1.006	65536	1.00
24	pair-b16-b128	36/36	13.25 ± 0.34	12.9	0.597	16384	0.25
24	pair-walk-b256	36/36	21.96 ± 0.17	21.4		65536	1.00
24	b8-klrand	36/36	1.54 ± 0.10	1.5	0.064	4233	0.13
24	b8-klprog	36/36	2.95 ± 0.27	2.9	0.112	9565	0.29
24	b8-null-strided	36/36	4.75 ± 0.16	4.6	0.212	16076	0.49
24	b8-hybrid	36/36	4.25 ± 0.17	4.1	0.187	14178	0.43
24	b8-nofold	36/36	6.77 ± 0.19	6.6	0.304	23341	0.71

24	b8-dp6	36/36	24.44 ± 0.95	23.8	1.087	1327	0.04
24	b8-dp4	36/36	11.60 ± 0.53	11.3	0.510	2519	0.08
24	b8-dp2	36/36	6.70 ± 0.26	6.5	0.295	5773	0.18
24	sweep-b128	36/36	13.28 ± 0.32	12.9	0.598	49476	1.51
24	sweep-b64	36/36	9.48 ± 0.21	9.2	0.428	34876	1.06
24	sweep-b32	36/36	7.15 ± 0.26	7.0	0.317	25835	0.79
24	sweep-b16	36/36	5.57 ± 0.16	5.4	0.250	19589	0.60
24	sweep-b8	36/36	4.23 ± 0.17	4.1	0.186	14178	0.43
24	sweep-b4	36/36	3.56 ± 0.17	3.5	0.153	11113	0.34
24	sweep-b2	36/36	2.93 ± 0.11	2.9	0.130	8065	0.25
24	kangaroo	36/36	3.20 ± 0.10	3.1	0.143	395	0.01
24	bsgs-neg	36/36	1.00 ± 0.05	1.0	0.044	1921	0.03
24	rho-lanes16	36/36	1.47 ± 0.08	1.4	0.063	0	0.00
24	rho-neg	36/36	1.03 ± 0.07	1.0	0.043	0	0.00
26	pair-b16-b768	36/36	75.81 ± 0.06	94.1	3.175	589824	9.00
26	pair-b16-b384	36/36	24.20 ± 0.19	30.1	1.013	147456	2.25
26	pair-b16-b192	36/36	16.27 ± 0.45	20.2	0.672	36864	0.56
26	pair-walk-b384	36/36	23.91 ± 0.23	29.7		147456	2.25
26	b8-klrand	36/36	1.54 ± 0.14	1.9	0.053	10285	0.31
26	b8-klprog	36/36	3.19 ± 0.31	4.0	0.113	23218	0.71
26	b8-null-strided	36/36	4.19 ± 0.16	5.2	0.171	30846	0.94
26	b8-hybrid	36/36	4.29 ± 0.16	5.3	0.175	31567	0.96
26	b8-nofold	36/36	5.59 ± 0.15	6.9	0.231	41574	1.27
26	b8-dp6	36/36	20.46 ± 0.83	25.4	0.826	2397	0.07
26	b8-dp4	36/36	10.54 ± 0.40	13.1	0.429	4951	0.15
26	b8-dp2	36/36	5.74 ± 0.22	7.1	0.233	10688	0.33
26	sweep-b128	36/36	11.81 ± 0.36	14.7	0.484	94332	2.88
26	sweep-b64	36/36	8.89 ± 0.15	11.0	0.370	70420	2.15
26	sweep-b32	36/36	6.63 ± 0.16	8.2	0.275	51803	1.58
26	sweep-b16	36/36	5.11 ± 0.11	6.3	0.212	39073	1.19
26	sweep-b8	36/36	4.28 ± 0.16	5.3	0.175	31567	0.96
26	sweep-b4	36/36	3.45 ± 0.14	4.3	0.140	23791	0.73
26	sweep-b2	36/36	3.16 ± 0.11	3.9	0.129	19577	0.60
26	kangaroo	36/36	0.87 ± 0.06	1.1	0.033	226	0.01
26	bsgs-neg	36/36	1.06 ± 0.00	1.3	0.044	4071	0.06
26	rho-lanes16	36/36	0.98 ± 0.09	1.2	0.033	0	0.00
26	rho-neg	36/36	0.81 ± 0.08	1.0	0.027	0	0.00
28	pair-b16-b1024	36/36	69.17 ± 0.11	67.2	2.790	1048576	16.00
28	pair-b16-b512	36/36	24.92 ± 0.26	24.2	1.003	262144	4.00
28	pair-b16-b256	36/36	21.52 ± 0.42	20.9	0.862	65536	1.00
28	pair-walk-b512	36/36	24.80 ± 0.11	24.1		262144	4.00
28	b8-klrand	36/36	1.49 ± 0.07	1.4	0.058	20970	0.64
28	b8-klprog	36/36	3.25 ± 0.23	3.2	0.121	47953	1.46
28	b8-null-strided	36/36	3.90 ± 0.11	3.8	0.155	57965	1.77
28	b8-hybrid	36/36	3.92 ± 0.11	3.8	0.156	58075	1.77
28	b8-nofold	36/36	4.70 ± 0.16	4.6	0.186	70051	2.14
28	b8-dp6	36/36	20.39 ± 0.96	19.8	0.788	4788	0.15
28	b8-dp4	36/36	11.34 ± 0.58	11.0	0.435	10694	0.33
28	b8-dp2	36/36	6.33 ± 0.23	6.1	0.250	23699	0.72
28	sweep-b128	36/36	11.70 ± 0.22	11.4	0.469	186529	5.69
28	sweep-b64	36/36	8.99 ± 0.21	8.7	0.359	142438	4.35
28	sweep-b32	36/36	6.49 ± 0.20	6.3	0.257	101637	3.10
28	sweep-b16	36/36	5.53 ± 0.07	5.4	0.223	85125	2.60

28	sweep-b8	36/36	3.91 ± 0.11	3.8	0.155	58075	1.77
28	sweep-b4	36/36	3.29 ± 0.14	3.2	0.129	45861	1.40
28	sweep-b2	36/36	2.59 ± 0.16	2.5	0.098	32331	0.99
28	kangaroo	36/36	2.72 ± 0.11	2.6	0.106	700	0.02
28	bsgs-neg	36/36	0.96 ± 0.06	0.9	0.036	8067	0.12
28	rho-lanes16	36/36	1.16 ± 0.08	1.1	0.042	0	0.00
28	rho-neg	36/36	1.03 ± 0.08	1.0	0.036	0	0.00
30	pair-b16-b1536	36/36	81.82 ± 0.16	112.2	2.764	2359296	36.00
30	pair-b16-b768	36/36	29.05 ± 0.48	39.8	0.976	589824	9.00
30	pair-b16-b384	36/36	24.22 ± 0.77	33.2	0.798	147456	2.25
30	pair-walk-b768	36/36	29.64 ± 0.25	40.7		589824	9.00
30	b8-klrand	36/36	1.44 ± 0.12	2.0	0.043	40222	1.23
30	b8-klprog	36/36	2.83 ± 0.26	3.9	0.082	80700	2.46
30	b8-null-strided	36/36	4.26 ± 0.10	5.8	0.142	122207	3.73
30	b8-hybrid	36/36	3.84 ± 0.09	5.3	0.129	110166	3.36
30	b8-nofold	36/36	5.01 ± 0.20	6.9	0.165	144089	4.40
30	b8-dp6	36/36	22.90 ± 0.85	31.4	0.757	10373	0.32
30	b8-dp4	36/36	9.67 ± 0.55	13.3	0.302	17480	0.53
30	b8-dp2	36/36	5.03 ± 0.22	6.9	0.164	36194	1.10
30	sweep-b128	36/36	11.81 ± 0.20	16.2	0.397	361405	11.03
30	sweep-b64	36/36	9.39 ± 0.17	12.9	0.315	285985	8.73
30	sweep-b32	36/36	6.75 ± 0.14	9.3	0.226	203671	6.22
30	sweep-b16	36/36	4.93 ± 0.15	6.8	0.164	146162	4.46
30	sweep-b8	36/36	3.84 ± 0.09	5.3	0.129	110166	3.36
30	sweep-b4	36/36	3.51 ± 0.13	4.8	0.115	95066	2.90
30	sweep-b2	36/36	2.75 ± 0.16	3.8	0.086	66590	2.03
30	kangaroo	36/36	1.76 ± 0.20	2.4	0.048	846	0.02
30	bsgs-neg	36/36	1.04 ± 0.03	1.4	0.034	15429	0.24
30	rho-lanes16	36/36	0.81 ± 0.07	1.1	0.023	0	0.00
30	rho-neg	36/36	0.73 ± 0.07	1.0	0.021	0	0.00
32	pair-b16-b2048	36/36	78.00 ± 0.27	76.5	2.366	4194304	64.00
32	pair-b16-b1024	36/36	31.08 ± 0.43	30.5	0.940	1048576	16.00
32	pair-b16-b512	36/36	34.52 ± 1.42	33.9	1.011	262144	4.00
32	pair-walk-b1024	36/36	33.01 ± 0.29	32.4		1048576	16.00
32	b8-klrand	36/36	1.48 ± 0.12	1.5	0.039	80372	2.45
32	b8-klprog	36/36	3.15 ± 0.22	3.1	0.088	173260	5.29
32	b8-null-strided	36/36	3.82 ± 0.15	3.8	0.113	210349	6.42
32	b8-hybrid	36/36	4.25 ± 0.12	4.2	0.127	233762	7.13
32	b8-nofold	36/36	5.70 ± 0.14	5.6	0.170	314291	9.59
32	b8-dp6	36/36	21.43 ± 1.00	21.0	0.625	18526	0.57
32	b8-dp4	36/36	11.73 ± 0.44	11.5	0.345	40487	1.24
32	b8-dp2	36/36	5.87 ± 0.20	5.8	0.175	80872	2.47
32	sweep-b128	36/36	11.55 ± 0.37	11.3	0.343	675642	20.62
32	sweep-b64	36/36	9.41 ± 0.25	9.2	0.282	548346	16.73
32	sweep-b32	36/36	6.80 ± 0.20	6.7	0.203	392384	11.97
32	sweep-b16	36/36	5.38 ± 0.13	5.3	0.162	305856	9.33
32	sweep-b8	36/36	4.25 ± 0.12	4.2	0.127	233762	7.13
32	sweep-b4	36/36	3.13 ± 0.13	3.1	0.092	162494	4.96
32	sweep-b2	36/36	2.82 ± 0.16	2.8	0.080	131732	4.02
32	kangaroo	36/36	1.70 ± 0.07	1.7	0.050	775	0.02
32	bsgs-neg	36/36	0.88 ± 0.06	0.9	0.025	29441	0.45
32	rho-lanes16	36/36	1.06 ± 0.09	1.0	0.028	0	0.00
32	rho-neg	36/36	1.02 ± 0.09	1.0	0.026	0	0.00

34	pair-b16-b1536	36/36	39.63 ± 0.38	49.6	0.993	2359296	36.00
34	pair-b16-b768	36/36	45.71 ± 1.59	57.3	1.111	589824	9.00
34	pair-walk-b1536	36/36	40.07 ± 0.67	50.2		2359296	36.00
34	b8-klrand	36/36	1.46 ± 0.11	1.8	0.033	165067	5.04
34	b8-klprog	36/36	2.48 ± 0.20	3.1	0.055	281693	8.60
34	b8-null-strided	36/36	4.12 ± 0.15	5.2	0.101	469567	14.33
34	b8-hybrid	36/36	3.95 ± 0.10	5.0	0.098	450715	13.75
34	b8-nofold	36/36	5.04 ± 0.23	6.3	0.122	574513	17.53
34	b8-dp4	36/36	10.91 ± 0.40	13.7	0.268	78038	2.38
34	b8-dp2	36/36	5.81 ± 0.11	7.3	0.145	165694	5.06
34	sweep-b128	36/36	11.83 ± 0.33	14.8	0.293	1430689	43.66
34	sweep-b64	36/36	8.90 ± 0.21	11.1	0.221	1071567	32.70
34	sweep-b32	36/36	6.11 ± 0.19	7.7	0.150	729395	22.26
34	sweep-b16	36/36	5.20 ± 0.11	6.5	0.129	610968	18.65
34	sweep-b8	36/36	3.95 ± 0.10	5.0	0.098	450715	13.75
34	sweep-b4	36/36	3.12 ± 0.13	3.9	0.076	336080	10.26
34	sweep-b2	36/36	3.29 ± 0.13	4.1	0.080	318301	9.71
34	kangaroo	36/36	1.53 ± 0.09	1.9	0.036	1446	0.03
34	bsgs-neg	36/36	0.97 ± 0.05	1.2	0.023	60778	0.93
34	rho-lanes16	36/36	0.82 ± 0.06	1.0	0.019	0	0.00
34	rho-neg	36/36	0.80 ± 0.05	1.0	0.018	0	0.00
36	pair-b16-b2048	36/36	44.31 ± 0.71	34.3	1.056	4194304	64.00
36	pair-b16-b1024	36/36	52.68 ± 1.41	40.7	1.246	1048576	16.00
36	pair-walk-b2048	36/36	42.17 ± 0.96	32.6		4194304	64.00
36	b8-klrand	36/36	1.16 ± 0.10	0.9	0.024	223557	6.82
36	b8-klprog	36/36	3.09 ± 0.29	2.4	0.064	599744	18.30
36	b8-null-strided	36/36	4.40 ± 0.18	3.4	0.102	853938	26.06
36	b8-hybrid	36/36	3.85 ± 0.10	3.0	0.091	748282	22.84
36	b8-nofold	36/36	5.17 ± 0.17	4.0	0.122	1004776	30.66
36	b8-dp4	36/36	10.63 ± 0.46	8.2	0.247	128932	3.93
36	b8-dp2	36/36	6.16 ± 0.23	4.8	0.144	299197	9.13
36	sweep-b128	36/36	11.89 ± 0.23	9.2	0.283	2447111	74.68
36	sweep-b64	36/36	8.99 ± 0.20	7.0	0.214	1842720	56.24
36	sweep-b32	36/36	6.38 ± 0.15	4.9	0.151	1297330	39.59
36	sweep-b16	36/36	5.26 ± 0.21	4.1	0.122	1052565	32.12
36	sweep-b8	36/36	3.85 ± 0.10	3.0	0.091	748282	22.84
36	sweep-b4	36/36	2.84 ± 0.12	2.2	0.066	521123	15.90
36	sweep-b2	36/36	2.28 ± 0.12	1.8	0.052	375346	11.45
36	kangaroo	36/36	1.97 ± 0.07	1.5	0.046	1562	0.04
36	bsgs-neg	36/36	1.29 ± 0.04	1.0	0.030	103364	1.58
36	rho-lanes16	36/36	1.33 ± 0.11	1.0	0.027	0	0.00
36	rho-neg	36/36	1.29 ± 0.11	1.0	0.026	0	0.00
38	rho-neg	12/12	0.96 ± 0.16	1.0		0	0.00
38	rho-lanes16	12/12	0.99 ± 0.15	1.0		0	0.00
38	bsgs-neg	12/12	1.28 ± 0.06	1.3		208430	3.18
38	kangaroo	12/12	1.45 ± 0.29	1.5		2358	0.05
38	sweep-b2	12/12	2.47 ± 0.19	2.6		822199	25.09
38	b2-dp4	12/12	7.20 ± 0.53	7.5		149965	4.58
38	b2-dp6	12/12	13.07 ± 0.94	13.6		68173	2.08
38	b8-dp4	12/12	10.17 ± 1.15	10.6		249048	7.60
38	b8-dp6	12/12	17.13 ± 2.20	17.8		105153	3.21
40	rho-neg	12/12	0.49 ± 0.06	1.0		0	0.00
40	rho-lanes16	12/12	0.51 ± 0.06	1.0		0	0.00

40	bsgs-neg	12/12	1.08 ± 0.01	2.2		418364	6.38
40	kangaroo	12/12	1.29 ± 0.13	2.6		2074	0.05
40	sweep-b2	12/12	2.40 ± 0.21	4.9		1602100	48.89
40	b2-dp4	12/12	7.23 ± 0.90	14.7		302328	9.23
40	b2-dp6	12/12	12.25 ± 0.76	24.8		127862	3.90
40	b8-dp4	12/12	7.83 ± 0.59	15.9		385161	11.75
40	b8-dp6	12/12	20.92 ± 1.66	42.4		257244	7.85
42	rho-neg	12/12	0.90 ± 0.12	1.0		0	0.00
42	rho-lanes16	12/12	0.91 ± 0.12	1.0		0	0.00
42	bsgs-neg	12/12	0.88 ± 0.00	1.0		878603	13.41
42	kangaroo	12/12	1.32 ± 0.02	1.5		4473	0.10
42	b2-dp4	12/12	7.11 ± 0.80	7.9		623948	19.04
42	b2-dp6	12/12	20.22 ± 0.93	22.5		443506	13.53
42	b8-dp4	12/12	13.73 ± 0.24	15.3		1419940	43.33
42	b8-dp6	12/12	17.17 ± 1.12	19.1		443785	13.54
44	rho-neg	12/12	1.28 ± 0.15	1.0		0	0.00
44	rho-lanes16	12/12	1.29 ± 0.15	1.0		0	0.00
44	bsgs-neg	12/12	0.82 ± 0.09	0.6		1491316	22.76
44	kangaroo	12/12	2.13 ± 0.09	1.7		6184	0.14
44	b2-dp4	12/12	8.22 ± 0.42	6.4		1225472	37.40
44	b2-dp6	12/12	14.24 ± 1.43	11.1		530644	16.19
44	b8-dp4	12/12	12.64 ± 0.75	9.9		2218790	67.71
44	b8-dp6	12/12	21.37 ± 2.23	16.7		937513	28.61